ADDITIONAL MATHEMATICS ASSIGNMENT

Sam Banankhu Mhango

088 472 0038

1 Prove that $y = x + \frac{1}{x}$ is a solution of the equation $\frac{dy}{dx} = 1 - 1x^2$.

2 Evaluate $\int \frac{(x^3+1)}{x^2}$

3 A particle start from rest and moves in astraight line with a speed given by $v = 4t - 2t^2$.Find how far it goes in the two seconds .

4 Evaluate $\int_1^2 x(x-1)(x-2)dx$

5 If $y = \frac{1}{\sqrt{1-t^2}}$ and $x = 3(1-t^3)$, find $\frac{dy}{dx}$ in terms of t .

6 If $y = 2t^2 - t$ and $x = 3t + 1$, find the value of $\frac{dy}{dt}$ and $\frac{dx}{dt}$.What is the Value of $\frac{dt}{dx}$?Deduce the value of $\frac{dy}{dx}$

7 At the point (x, y)on the curve, the gradient is equal to $(2x - 1)$.If the Curve pass through the point $(3, 4)$ what is the equation?

8 A body moves along a line OA, so that its velocity t seconds after passing O is $2 + 3t^2$ ms^{-1}. T he time from O to A IS 2 seconds.Find the length OA.

9 Given thet $\frac{d^2y}{dx^2} = 4x$,and that $\frac{dy}{dx} = 0$,$y = 2$ when $x = 0$. Find $\frac{dy}{dx}$ and y in terms of x.

10 If the fisrt three terms of the expassion $(a + 3x)^n$ are $b + \frac{21}{2}bx + \frac{189}{4}bx^2$,

Find the value of a, b *and* n

a) In the expassion of $(2p - q)^4$, find the coefficient of p^3q
b) The coefficient x^3 in the expassion of $(1 + kx)^{10}$ is 32.Find the value of k
c) Find the coefficient of the term with x^{10} in the expassion of $(2x + y)^{14}$

11 Find the term independent of $\boldsymbol{x}$ in the expassion of $\left(\boldsymbol{x}+\frac{\mathbf{1}}{\mathbf{2}\boldsymbol{x}}\right)^{8.}$

12 Expand $(\mathbf{1}+\boldsymbol{x})^6$ and in the result substitute $\boldsymbol{x}=\mathbf{0.01}$.Hence obtain the

value Of $(\mathbf{1.01})^6$ correct to 5 decimal places

13 Find the maximum and minimum value of the function $2\sin x \ +\cos x\ (x\ in\ radians)$

14 Find the point of intersection of the tangent to the curve $y=2-3x+x^2$

15 A cylindrical can $(with\ a\ lid)$ of radius r cm is made from 300cm^2 of thin sheet

Metal

a) Show that its height h cm is given by $h=\frac{150-\pi r^2}{\pi r}$.
b) Find r and h so that the can will contain maximum possible volume and find its volume.

16 A rectangular box with a lid is made from thin metal , its length $l=2x\ cm$ and its width

$w=x\ cm$ .If the box must have a volume of $72cm^3$

a) Show that the area Acm^2 of the metal is given by $A=4x^2+\frac{216}{x}$.
b) Find the value of X so that A is a maximum.

17 A piece of a wire 20cm is formed into shape of a sector of a circle radius r cm and angle

$\theta\ radian$.

a) Show that $\theta=\frac{20-2r}{r}$ and that the area of a sector is $r(10-r)\ \ cm^2$
b) Hence find the value of $r\ \ and\ \ \theta$ to give the maximum area.

18 A particle moves along the x axis and its x coordinate at time Ts after the start is given by

$x=3\,t^3-9t^2\ +12t-1$ for the $t\geq 0$.

a) Find its X coordinate and velocity at the start
b) At what time does the particle come to instantaneous rest
c) What is it maximum velocity in the directon of the negative X axis
d) When its acceleration is zero

19 The velocity $v\ cm^{-1}$ of a particle moving in a straight line is given by$v=6t-kt^2$ where K

Is constant and T is the time from the start.If its acceleration is 0 when t=1, find

a) The value of K
b) The time when the particle comes to instanteneous rest
c) The maximum velocity of the particle

20 For a particle moving in a straight line, its displacement S m from a fixed point O is given by

$s = t^2 - 5t + 6$, where T is time in seconds from the start. Find

a) Its initial distance of the particle from O
b) Its initial velocity
c) When is it at instantaneous rest
d) The distance travelled in the fist 3 seconds

21 The distance S m of a particle moving from fixed position O on the line given by

$s = t^2 - 2t$ where T is the time in seconds after the start.

a) What is the initial velocity of the particle
b) When is the particle at instantaneous rest
c) When does it pass through point O for the second time
d) What is the acceleration of the particle

22 A small body moves along the X axis so that its distance X from original at time T is given by

$x = 2t^3 - 15t^2 + 24t + 20$.Find

a) The velocity with which it start
b) When it is at instantaneous rest
c) The maximum distance of the body from the origin
d) Between what time is the particle moving toward the origin
e) What is its acceleration at the time in D

23 A particle moves in straight line its displacement is given S m from fixed point on t he line is

given by $s = t^2 - 4t - 5$,at time T after start where $t \geq 0$.Find

a) Where the particle start and its initial velocity
b) When where it comes to instantaneous rest
c) When it pass through the fixed point
d) Its acceleration

24 Given that $l = \frac{1}{a+bx}$ and that L=1 and $\frac{dL}{dx} = 3$b when x=1.Find the value of a and b

25 The curve $y = \frac{1}{a+bx}$ passes through point $(1, -1)$ and its gradient at that point is 2
$Find$ $the\ value$ of a and $b.$

26 Use the binomial theorem to find the first three terms in the expassion of $\left[1-\left(y-\frac{y^2}{4}\right)\right]^4$

In form of $a+by+cy^2$.

27 The term with x^5 in the expassion $(1+3x)^{10}$ is equal to the term with x^3 in the expassion

Of $(252+x)^9$ .Find the value of X

28 Find the value of a and n if the sum of the first three terms in the expassion of $(1+ax)^n$

Is $1+16x+96x^2$.

29 If the first three terms of the expassion of $(a+3x)^n$ are $b+\frac{21}{2}bx+\frac{189}{4}bx^2$, find the value

Of a , b and n.

a) Obtain the expassion of $(\alpha-x)^5$ .If $\left(\sqrt{5}-\sqrt{3}\right)^5=A\sqrt{5}-B\sqrt{3}$, Find the value of A and B
b) Find the coefficient of x^4 in the expassion of $(1+x+2x^2)^{0.5}$

30 Given that $y=3x^3$, find the approximate % increase in y due to an increase of 0.1% in X.